

BUSFIN 8200: Problem Set 1

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1 Question 1

1.1 Part (a)

Assume $D_t > 0$. By definition, the gross return on equity is

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}. \quad (1)$$

Taking logs,

$$\begin{aligned} \log(R_{e,t+1}) &= \log\left(\frac{P_{t+1} + D_{t+1}}{P_t}\right) \\ &= \log\left(\frac{P_{t+1} + D_{t+1}}{P_{t+1}} \cdot P_{t+1}\right) - \log(P_t) \\ &= \log\left[\left(1 + \frac{D_{t+1}}{P_{t+1}}\right) P_{t+1}\right] - \log(P_t) \\ &= \log(P_{t+1}) + \log\left(1 + \frac{D_{t+1}}{P_{t+1}}\right) - \log(P_t). \end{aligned}$$

Define $r_{e,t+1} \equiv \log(R_{e,t+1})$, $p_t \equiv \log(P_t)$, and $d_t \equiv \log(D_t)$. Since $\frac{D_{t+1}}{P_{t+1}} = e^{\log(D_{t+1}/P_{t+1})} = e^{d_{t+1}-p_{t+1}}$, we can write

$$r_{e,t+1} = p_{t+1} + \log\left(1 + e^{d_{t+1}-p_{t+1}}\right) - p_t.$$

Define $f(x) \equiv \log(1 + e^x)$, where $x = d_{t+1} - p_{t+1}$. We approximate this function around the average log dividend-price ratio $\bar{d} - \bar{p}$ using a first-order Taylor approximation,

$$f(x) \approx f(\bar{d} - \bar{p}) + f'(\bar{d} - \bar{p})(x - \bar{d} - \bar{p}),$$

so that

$$f(d_{t+1} - p_{t+1}) \approx f(\bar{d} - \bar{p}) + f'(\bar{d} - \bar{p})[(d_{t+1} - p_{t+1}) - \bar{d} - \bar{p}].$$

Since $f(x) = \log(1 + e^x)$, we have $f'(x) = \frac{e^x}{1 + e^x}$, so

$$f'(\bar{d} - \bar{p}) = \frac{e^{\bar{d} - \bar{p}}}{1 + e^{\bar{d} - \bar{p}}}.$$

Define $\kappa \equiv \frac{1}{1 + e^{\bar{d} - \bar{p}}}$. Then $1 - \kappa = \frac{e^{\bar{d} - \bar{p}}}{1 + e^{\bar{d} - \bar{p}}}$, so $f'(\bar{d} - \bar{p}) = 1 - \kappa$.

Hence,

$$\begin{aligned} f(d_{t+1} - p_{t+1}) &\approx f(\overline{d-p}) + (1-\kappa) \left[(d_{t+1} - p_{t+1}) - \overline{d-p} \right] \\ &= f(\overline{d-p}) - (1-\kappa)\overline{d-p} + (1-\kappa)(d_{t+1} - p_{t+1}). \end{aligned}$$

Define $\kappa_0 \equiv f(\overline{d-p}) - (1-\kappa)\overline{d-p}$, a constant. Since $f(\overline{d-p}) = \log(1+e^{\overline{d-p}})$ and $\kappa = \frac{1}{1+e^{\overline{d-p}}}$, we have $1+e^{\overline{d-p}} = \frac{1}{\kappa}$, so

$$f(\overline{d-p}) = \log\left(\frac{1}{\kappa}\right) = -\log \kappa, \quad e^{\overline{d-p}} = \frac{1}{\kappa} - 1, \quad \overline{d-p} = \log\left(\frac{1}{\kappa} - 1\right).$$

Therefore,

$$\kappa_0 \equiv f(\overline{d-p}) - (1-\kappa)\overline{d-p} = -\log \kappa - (1-\kappa) \log\left(\frac{1}{\kappa} - 1\right).$$

We then have $\log(1+e^{d_{t+1}-p_{t+1}}) = f(d_{t+1} - p_{t+1}) \approx \kappa_0 + (1-\kappa)(d_{t+1} - p_{t+1})$, so

$$\begin{aligned} r_{e,t+1} &= p_{t+1} + \log\left(1+e^{d_{t+1}-p_{t+1}}\right) - p_t \\ &\approx p_{t+1} + \kappa_0 + (1-\kappa)(d_{t+1} - p_{t+1}) - p_t \\ &= \kappa_0 + (1-\kappa)d_{t+1} + \kappa p_{t+1} - p_t \\ &= \kappa_0 + \left[(1-\kappa)d_{t+1} - d_t + \kappa p_{t+1} - p_t + d_t\right] \\ &= \kappa_0 + (d_{t+1} - d_t) + \kappa(p_{t+1} - d_{t+1}) - (p_t - d_t) \\ &= \kappa_0 + \Delta d_{t+1} + \kappa(p_{t+1} - d_{t+1}) - (p_t - d_t). \end{aligned}$$

Therefore, we have

$$\begin{aligned} r_{e,t+1} &\approx \kappa_0 + \Delta d_{t+1} + \kappa(p_{t+1} - d_{t+1}) - (p_t - d_t), \\ \kappa &= \frac{1}{1+e^{\overline{d-p}}}, \\ \kappa_0 &= -\log(\kappa) - (1-\kappa) \log\left(\frac{1}{\kappa} - 1\right). \end{aligned}$$

Isolating $p_t - d_t$ and recursively substituting for $p_{t+h} - d_{t+h}$, we have

$$\begin{aligned} (p_t - d_t) &\approx \kappa_0 - r_{e,t+1} + \Delta d_{t+1} + \kappa(p_{t+1} - d_{t+1}) \\ &= \kappa_0 - r_{e,t+1} + \Delta d_{t+1} + \kappa \left[\kappa_0 - r_{e,t+2} + \Delta d_{t+2} + \kappa(p_{t+2} - d_{t+2}) \right] \\ &= (1+\kappa)\kappa_0 + (\Delta d_{t+1} + \kappa \Delta d_{t+2}) - (r_{e,t+1} + \kappa r_{e,t+2}) + \kappa^2(p_{t+2} - d_{t+2}) \\ &\vdots \text{ keep substituting } p_{t+h} - d_{t+h} \\ &= \frac{\kappa_0(1-\kappa^H)}{1-\kappa} + \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} - \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} + \kappa^H(p_{t+H} - d_{t+H}). \end{aligned}$$

Thus, we have

$$d_t - p_t \approx \frac{\kappa_0(\kappa^H - 1)}{1-\kappa} + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H(d_{t+H} - p_{t+H}),$$

so, letting $dp_t \equiv d_t - p_t$,

$$dp_t \approx \frac{\kappa_0(\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H dp_{t+H}. \quad (1.1)$$

Taking expectations as of time t ,

$$dp_t = \frac{\kappa_0(\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] + \kappa^H \mathbb{E}_t[dp_{t+H}]. \quad (1.2)$$

Suppose $H \rightarrow \infty$. Since $\kappa = \frac{1}{1 + e^{\bar{d}-p}} \in (0, 1)$, we have $\lim_{H \rightarrow \infty} \kappa^H = 0$, and by the no-bubble condition $\lim_{H \rightarrow \infty} \kappa^H \mathbb{E}_t[dp_{t+H}] = 0$. Therefore,

$$dp_t = \frac{-\kappa_0}{1 - \kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}]. \quad (1.3)$$

1.2 Part (b)



Figure 1: OLS slope coefficients of `future_return`, `change_d`, and `dp_H` on `dp`, plotted against the horizon H (in years).

From the equation

$$1 = \underbrace{\frac{\text{Cov} \left[dp_t, \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} \right]}{\text{Var}[dp]}}_{b_{re}^{(H)}} + \underbrace{\frac{\text{Cov} \left[dp_t, -\sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right]}{\text{Var}[dp]}}_{b_{\Delta d}^{(H)}} + \underbrace{\frac{\text{Cov} [dp_t, \kappa^H dp_{t+H}]}{\text{Var}[dp]}}_{b_{dp}^{(H)}}.$$

This equation is trying to measure, how much each component moves when today's dp_t moves. The meaning of these three coefficients is as follows:

$$\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] = a + b_{r_e}^{(H)} dp_t + \epsilon_t$$

The slope is $b_{r_e}^{(H)}$, and it measures the sensitivity of expected discounted future returns to today's dividend-price ratio.

Because dividend growth enters the Campbell–Shiller equation with a minus sign, define

$$-\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] = a + b_{\Delta d}^{(H)} dp_t + \epsilon_t.$$

So $b_{\Delta d}^{(H)}$ captures how much variation in dp_t is associated with expected future dividend growth.

$$\kappa^H \mathbb{E}_t[dp_{t+H}] = a + b_{dp}^{(H)} dp_t + \epsilon_t.$$

$b_{dp}^{(H)}$ tells us how much today's high dp_t is simply expected to persist into the future.

$$1 = b_{r_e}^{(H)} + b_{\Delta d}^{(H)} + b_{dp}^{(H)}$$

From the graph, we can see that the contributions of expected future returns, expected future dividend growth, and the future dividend-price ratio to the variation in today's dp_t change with the horizon H . In the short run, the variation in today's dp_t is primarily driven by the future dividend-price ratio. As the horizon H increases, the contribution of expected future returns becomes more significant, while the contribution of the future dividend-price ratio declines toward zero. The contribution of negative expected future dividend growth remains relatively stable across different horizons. Also, if we add up these three coefficients, we get approximately 1, which reflects the normalized variance decomposition of today's dp_t .

1.3 Part (c)



Figure 2: VAR-implied coefficients

The VAR-implied coefficients shown in Figure 2 show some interesting patterns. Instead of the expected future return component alone becoming dominant as H increases, the figure shows that both the expected future return component and the negative expected future dividend growth component contribute significantly to the variation in today's dp_t across different horizons. At the same time, the contribution of the future dividend-price ratio declines as the horizon increases, consistent with the findings from the direct-regression estimates, although the decline appears to be more gradual under the VAR approach. The difference in these patterns may arise because the two approaches estimate long-horizon coefficients differently. The direct approach estimates each horizon separately using realized future outcomes, whereas the VAR approach estimates the joint one-year dynamics of the state variables and uses the estimated transition matrix to imply coefficients at all subsequent horizons.

1.4 Part (d)

The resulting long-run VAR-implied coefficients are reported in Table 1. The table shows that, in the long run, the contributions of expected future returns and negative expected future dividend growth are roughly equal, accounting for approximately 49% and 51%, respectively. Their sum is 0.998808, which is close to, but not exactly, one. This indicates that the unrestricted estimated VAR approximately, rather than mechanically, satisfies the long-run Campbell–Shiller restriction.

Table 1: Long-run VAR-implied coefficients

Component	Coefficient	Estimate
Expected return	$b_{re}^{(\infty)}$	0.489191
Negative expected dividend growth	$b_{dg}^{(\infty)}$	0.509617
Sum	$b_{re}^{(\infty)} + b_{dg}^{(\infty)}$	0.998808

1.5 Part (e)

First, assume $D_t > 0$. We have

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}.$$

Then,

$$\begin{aligned} R_{e,t+1} \frac{D_t}{D_{t+1}} &= \frac{P_{t+1} + D_{t+1}}{P_t} \frac{D_t}{D_{t+1}} \\ &= \frac{D_t}{P_t} \frac{P_{t+1} + D_{t+1}}{D_{t+1}} \\ &= \frac{P_t + D_t}{P_t} \frac{D_t}{P_t + D_t} \\ &= \frac{D_t}{P_{t+1} + D_{t+1}}. \end{aligned}$$

Taking logs on both sides gives

$$\begin{aligned} \log(R_{e,t+1}) + \log\left(\frac{D_t}{D_{t+1}}\right) &= \log\left(\frac{P_t + D_t}{P_t}\right) + \log\left(\frac{D_t}{P_t + D_t}\right) \\ &\quad - \log\left(\frac{D_{t+1}}{P_{t+1} + D_{t+1}}\right). \end{aligned}$$

Define

$$r_{e,t+1} = \log(R_{e,t+1}), \quad p_t = \log(P_t), \quad d_t = \log(D_t), \quad dy_t = \log\left(1 + \frac{D_t}{P_t}\right).$$

Given that

$$\begin{aligned} \log\left(\frac{D_t}{P_t + D_t}\right) &= \log\left(1 - \frac{P_t}{P_t + D_t}\right) \\ &= \log\left(1 - e^{\log(P_t)/(P_t + D_t)}\right) \\ &= \log\left(1 - e^{-\log((P_t + D_t)/P_t)}\right) \\ &= \log\left(1 - e^{-\log(1 + D_t/P_t)}\right) \\ &= \log\left(1 - e^{-dy_t}\right), \end{aligned}$$

we have

$$\log\left(\frac{D_t}{P_t + D_t}\right) = \log\left(1 - e^{-dy_t}\right).$$

Similarly,

$$\log\left(\frac{D_{t+1}}{P_{t+1} + D_{t+1}}\right) = \log\left(1 - e^{-dy_{t+1}}\right).$$

Therefore,

$$r_{e,t+1} - \Delta d_{t+1} = dy_t + \log\left(1 - e^{-dy_t}\right) - \log\left(1 - e^{-dy_{t+1}}\right).$$

Let

$$f(x) = \log(1 - e^{-x}).$$

Then,

$$f'(x) = \frac{1}{1 - e^{-x}} e^{-x} = \frac{e^{-x}}{1 - e^{-x}}.$$

Let $x = dy_t$. By a first-order approximation, and defining $\kappa \equiv e^{-\bar{dy}}$, we have

$$\begin{aligned} \log(1 - e^{-dy_t}) &\approx \log(1 - e^{-\bar{dy}}) + \log'(1 - e^{-\bar{dy}})(dy_t - \bar{dy}) \\ &= \log(1 - \kappa) + \frac{e^{-\bar{dy}}}{1 - e^{-\bar{dy}}}(dy_t - \bar{dy}) \\ &= \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \bar{dy}). \end{aligned}$$

Thus,

$$\log(1 - e^{-dy_t}) \approx \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \bar{dy}).$$

Similarly,

$$\log(1 - e^{-dy_{t+1}}) \approx \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_{t+1} - \bar{dy}).$$

Substituting these approximations into $r_{e,t+1} - \Delta d_{t+1} = dy_t + \log(1 - e^{-dy_t}) - \log(1 - e^{-dy_{t+1}})$, we obtain

$$\begin{aligned} r_{e,t+1} - \Delta d_{t+1} &= dy_t + \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \bar{dy}) \\ &\quad - \log(1 - \kappa) - \frac{\kappa}{1 - \kappa}(dy_{t+1} - \bar{dy}) \\ &= dy_t + \frac{\kappa}{1 - \kappa}(dy_t - dy_{t+1}) \\ &= \frac{1}{1 - \kappa}dy_t - \frac{\kappa}{1 - \kappa}dy_{t+1}. \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{1}{1 - \kappa}dy_t &= r_{e,t+1} - \Delta d_{t+1} + \frac{\kappa}{1 - \kappa}dy_{t+1}, \\ dy_t &= (1 - \kappa)(r_{e,t+1} - \Delta d_{t+1}) + \kappa dy_{t+1}. \end{aligned}$$

Recursively substituting for future values of dy , we have

$$\begin{aligned} dy_t &= (1 - \kappa)(r_{e,t+1} - \Delta d_{t+1}) \\ &\quad + \kappa[(1 - \kappa)(r_{e,t+2} - \Delta d_{t+2}) + \kappa dy_{t+2}] \\ &\quad \vdots \quad \text{keep substituting } dy_{t+h} \\ &= (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right) + \kappa^H dy_{t+H}. \end{aligned}$$

Thus, we have

$$dy_t = (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right) + \kappa^H dy_{t+H}.$$

Taking expectations at time t , we have

$$dy_t = (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] \right) + \kappa^H \mathbb{E}_t[dy_{t+H}].$$

Suppose $H \rightarrow \infty$. Given that $\kappa = e^{-\bar{d}y} \in (0, 1)$, we have

$$\lim_{H \rightarrow \infty} \kappa^H = 0.$$

Also, because of the no-bubble condition,

$$\lim_{H \rightarrow \infty} \kappa^H \mathbb{E}_t[dy_{t+H}] = 0.$$

Therefore,

$$dy_t = (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] \right).$$

2 Question 2

2.1 Part (a)

Figure 3 shows that the adjusted R^2 generally rises with the horizon, from about 4.6% at one year to a peak of about 44.3% at 14 years, before declining slightly to about 42.1% at 15 years. Thus, the current dividend-price ratio has substantially greater explanatory power for long-horizon average excess returns than for one-year-ahead excess returns.



Figure 3: Adjusted R^2 from predictive regressions of average future annual simple excess returns on the starting dividend-price ratio.

2.2 Part (b)

Table 2: Q2(b) slope estimate and alternative standard errors

Method	$\hat{b}$	$\text{SE}(\hat{b})$	t -statistic	Lags/bandwidth
Baseline OLS	2.8038	0.3802	7.3754	0
White (HC0)	2.8038	0.6587	4.2569	0
Newey–West (1987)	2.8038	1.2990	2.1584	11
Hansen–Hodrick (1980)	2.8038	1.4537	1.9287	11
Newey–West (1994), automatic	2.8038	1.2523	2.2390	22.0776 ($L = 22$)

2.3 Part (c)

From Table 3, the Amihud and Hurvich estimate of b is 2.3412, which is smaller than the standard OLS estimate of 2.8038. It is natural for the two estimates to differ because as indicated by the estimated persistence coefficient $\hat{\phi} = 0.7197$, the predictor D_t/P_t is persistent. In a finite sample, persistence in the predictor can induce small-sample bias in the OLS estimate of the predictive-regression coefficient. The Amihud and Hurvich procedure corrects for this problem by first bias-correcting the persistence coefficient from $\hat{\phi} = 0.7197$ to $\hat{\phi}^c = 0.7540$ and then including the resulting bias-corrected residual $\hat{u}_{t+1}^c$ in the predictive regression. Thus, the difference between $\hat{b}_{\text{AH}}$ and $\hat{b}_{\text{OLS}}$ is expected.

Table 3: Q2(c) Amihud–Hurvich estimates

Quantity	Estimate
$\hat{\theta}$	0.0111
$\hat{\phi}$	0.7197
T (years)	95
$\hat{\phi}^c$	0.7540
$\hat{b}_{\text{AH}}$	2.3412
$\hat{b}_u$	−13.4837
$\hat{b}_{\text{OLS}}$	2.8038
$\hat{b}_{\text{AH}} - \hat{b}_{\text{OLS}}$	−0.4626
Observations	1,117

2.4 Part (d)

I estimate Equation (2.2) out of sample using an expanding window. The first out-of-sample forecast corresponds to December 1940, using the dividend-price ratio observed in December 1939. At each forecast date, I re-estimate the predictive regression using the historical information available at that date and use the resulting coefficients to construct the out-of-sample forecast. I also calculate the historical expanding-window mean excess return, $\overline{xR}_{e,t}$, which serves as the benchmark forecast. The in-sample fitted values are obtained from the full-sample regression in Equation (2.2).

The out-of-sample R^2 is calculated as

$$R_{OS}^2 = 1 - \frac{\sum_t \left(xR_{e,t+1} - \hat{E}_t^{OS}[xR_e] \right)^2}{\sum_t \left(xR_{e,t+1} - \overline{xR}_{e,t} \right)^2}.$$

The R_{OS}^2 compares the forecast errors from the predictive regression with those from the historical expanding-window mean benchmark. A positive R_{OS}^2 indicates that the dividend-price-ratio forecast has a lower sum of squared forecast errors than the historical-mean forecast, whereas a negative value indicates that the historical mean performs better.

For the full out-of-sample period, I obtain $R_{OS}^2 = -0.0061$. Thus, over the full evaluation period, the dividend-price-ratio forecast performs slightly worse than the historical-mean benchmark.

To examine variation in forecasting performance over time, I also calculate R_{OS}^2 using a 50-year (600-month) rolling window of forecast errors. The coefficients a_τ and b_τ continue to be estimated using the expanding window. Thus, the 50-year rolling window changes only the forecast errors included in the calculation of R_{OS}^2 ; it does not change how the forecasts themselves are constructed.

Figure 5 shows substantial variation in out-of-sample forecasting performance over time. The rolling R_{OS}^2 is strongly positive in the early part of the sample, indicating that the dividend-price-ratio forecast outperforms the historical-mean benchmark over those 50-year windows. It subsequently declines and fluctuates around zero for much of the later sample, before becoming increasingly negative toward the end.

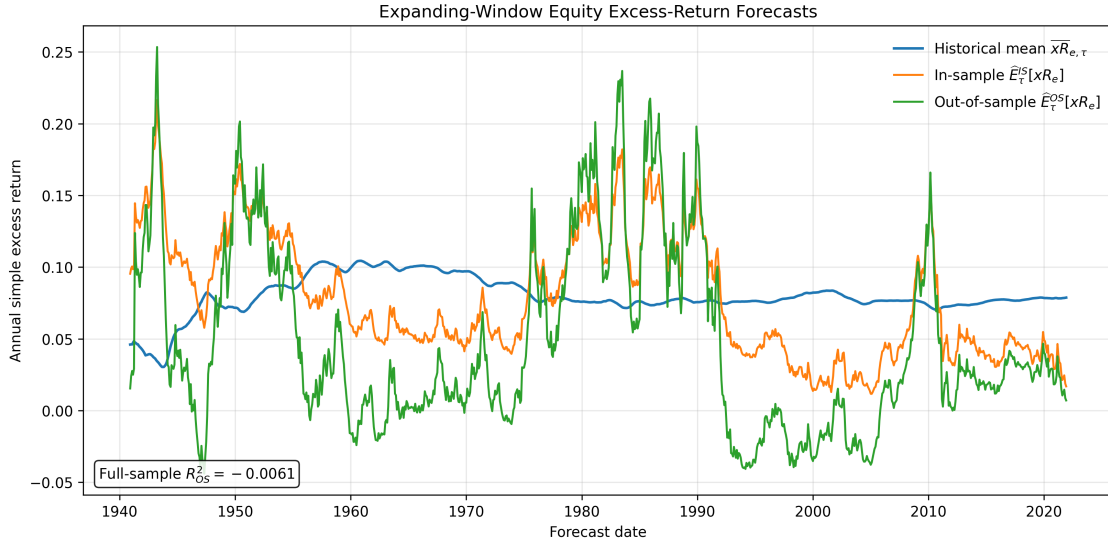


Figure 4: Historical-mean, in-sample, and expanding-window out-of-sample forecasts of the annual simple equity excess return.



Figure 5: Fifty-year rolling out-of-sample R^2 .

2.5 Part (e)

I repeat the out-of-sample exercise in Part (d), but impose the steady-state restrictions on the forecasting coefficients. For each forecast date τ , I calculate

$$G_\tau = \text{mean} \left(e^{dgs} \right)$$

using the same expanding historical sample used to calculate the historical-mean forecast $\overline{xR}_{e,\tau}$. I then impose

$$\hat{a}_\tau = G_\tau - 1, \quad \hat{b}_\tau = G_\tau,$$

so that the restricted out-of-sample forecast is

$$\hat{E}_\tau^{OS}[xR_e] = (G_\tau - 1) + G_\tau DP_\tau.$$

Thus, unlike Part (d), the coefficients are not estimated by OLS at each forecast date; instead, they are determined by the steady-state valuation restrictions.

Figure 6 compares the restricted out-of-sample forecast with the historical-mean forecast and the full-sample in-sample fitted values. Over the full out-of-sample evaluation period, the restricted forecast produces

$$R_{OS}^2 = 0.0321.$$

Since this value is positive, the restricted steady-state forecast has a lower sum of squared forecast errors than the historical-mean benchmark over the full out-of-sample period.

Figure 7 reports the 50-year rolling R_{OS}^2 . As in Part (d), the rolling window is applied only to the forecast errors used to calculate R_{OS}^2 ; G_τ continues to be calculated using the expanding historical window. The rolling R_{OS}^2 remains positive throughout the plotted period, indicating that the restricted forecast outperforms the historical-mean benchmark over each of these 50-year evaluation windows. However, its magnitude varies over time and declines substantially toward the end of the sample, indicating that the degree of out-of-sample improvement is not constant over time.

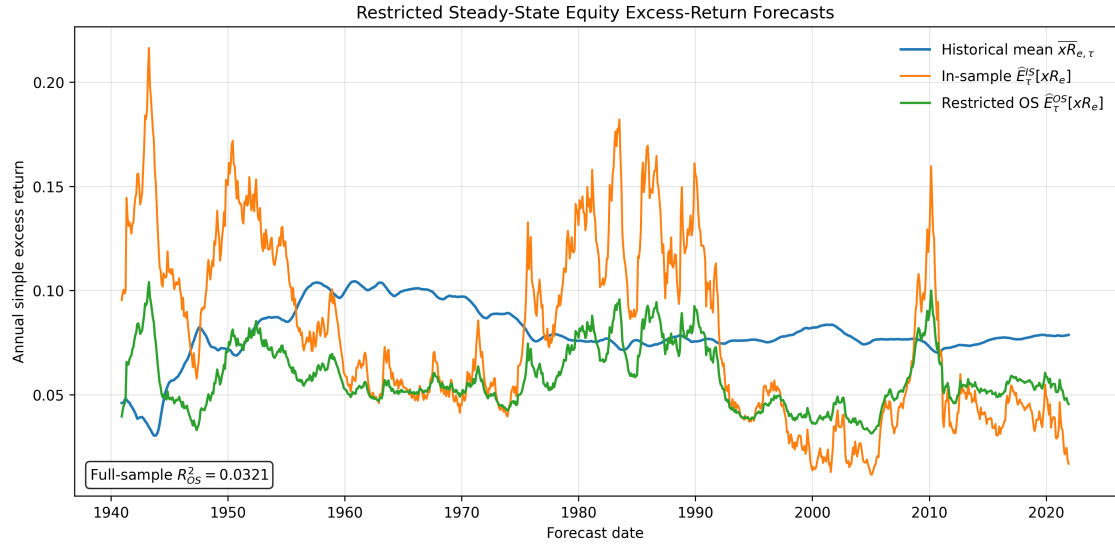


Figure 6: Historical-mean, in-sample, and restricted steady-state out-of-sample forecasts of the annual simple equity excess return.

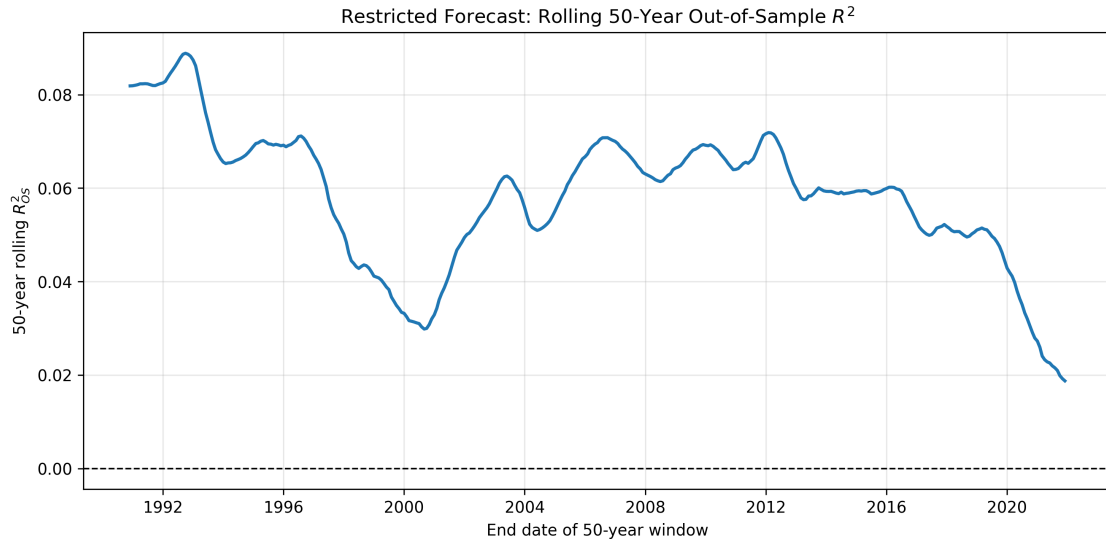


Figure 7: Fifty-year rolling out-of-sample R^2 for the restricted forecast.

3 Question 3

3.1 Part (a)

I construct the monthly momentum signal for each stock using CRSP returns. Momentum at the end of month τ is the cumulative return over the preceding 12 months, excluding the current month:

$$MOM_{i,\tau} = \prod_{k=1}^{12} (1 + RET_{i,\tau-k}) - 1$$

I require 12 valid monthly returns in consecutive calendar months. The sample begins in June 1963 and is restricted to common stocks with share codes 10 or 11 listed on NYSE, Amex or Nasdaq. I exclude utilities and financial firms.

To validate the constructed momentum signal, I merge the CRSP observations with the Chen–Zimmermann dataset. I then estimate the following cross-sectional regression separately for each month:

$$MOMCZ_{i,\tau} = a_{\tau} + b_{\tau}MOM_{i,\tau} + \varepsilon_{i,\tau}.$$

Figures 8–10 plot the monthly intercept estimates, slope estimates and (R^2) values, respectively.

The estimated intercepts generally fluctuate around zero. The slopes generally remain close to one. The (R^2) values are consistently high and close to one. The results suggest that the CRSP-based momentum signal closely matches the Chen–Zimmermann measure.

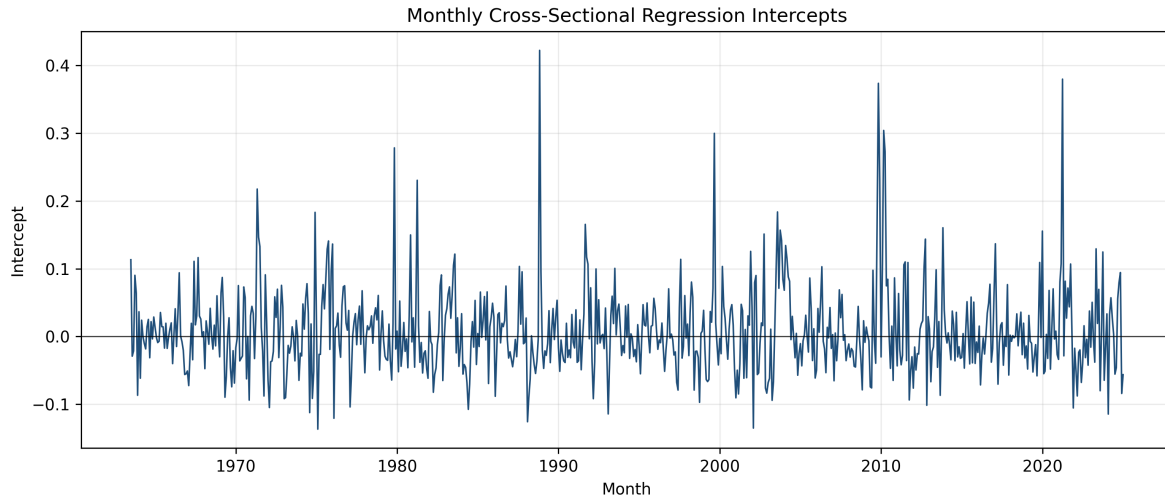


Figure 8: Monthly cross-sectional regression intercepts.



Figure 9: Monthly cross-sectional regression slopes.



Figure 10: Monthly cross-sectional regression R^2 values.

3.2 Part (b)

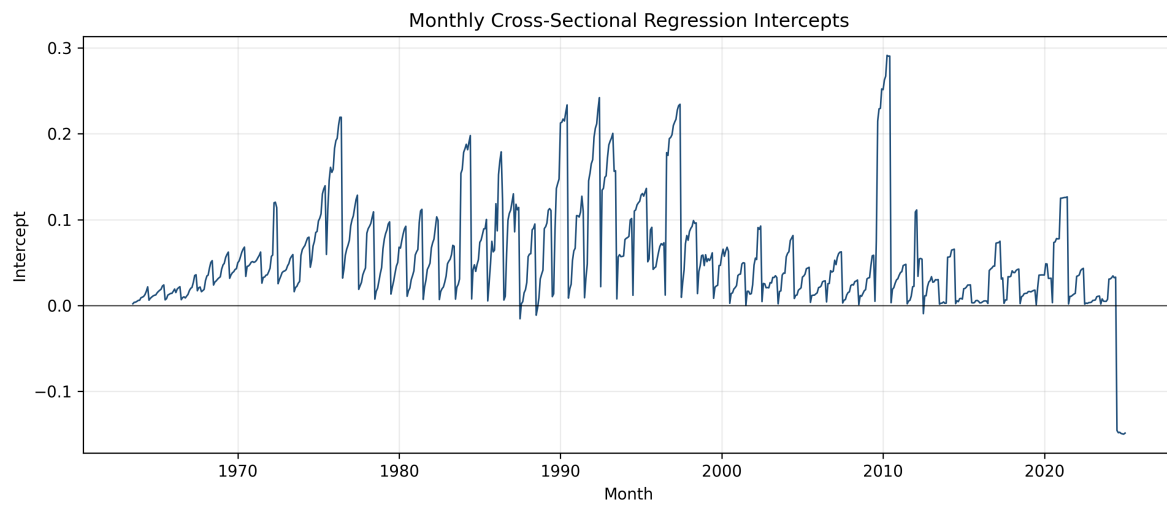


Figure 11: Monthly cross-sectional regression intercepts.

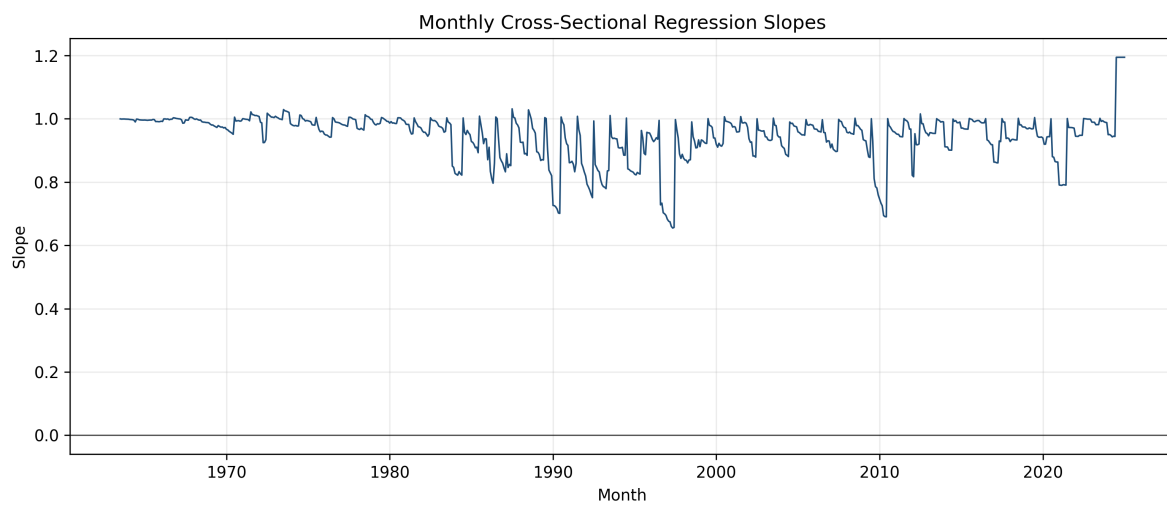


Figure 12: Monthly cross-sectional regression slopes.

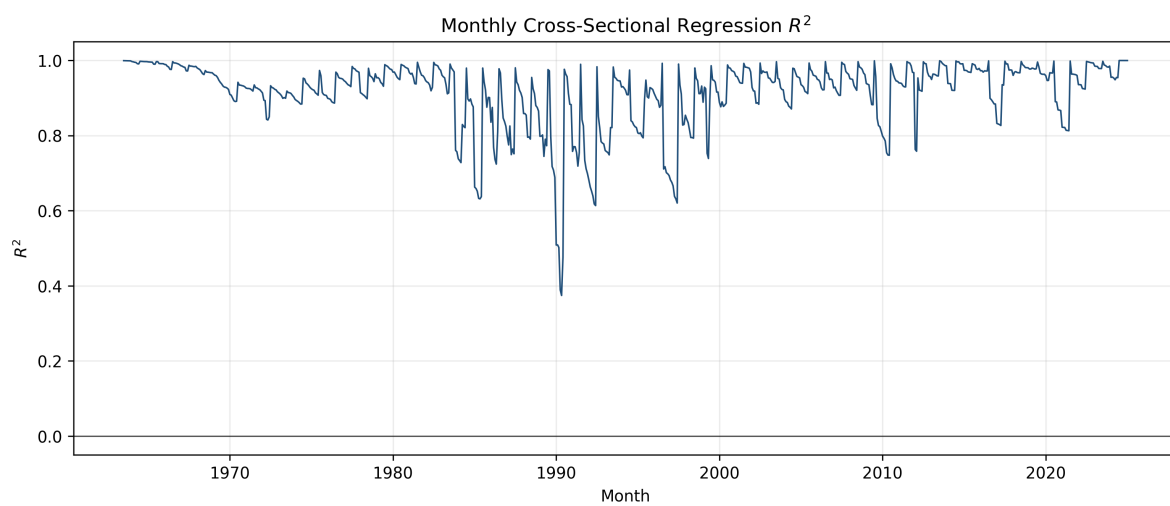


Figure 13: Monthly cross-sectional regression R^2 values.

4 Question 4

4.1 Part (a)

I use the Fama–Bliss yields for maturities from one to five years. Since the original yields are reported in percentage terms, I first divide them by 100 and then construct the log yields, log forward rates, and log annual returns according to the formulas in the question. Because the data are monthly, I use a 12-month lag when constructing annual returns.

For each maturity $H = 2, 3, 4, 5$, I calculate the excess log yield, excess log forward rate, and excess log annual return relative to the corresponding one-year value. Table 4 reports their time-series averages in percentage terms, obtained by multiplying the averages in log units by 100.

Table 4: Time-series averages of bond excess log yields, forward rates, and annual returns (percent)

H	$\overline{xy}_b^{(H)}$	$\overline{xf}_b^{(H)}$	$\overline{xr}_b^{(H)}$
2	0.168613	0.337226	0.315423
3	0.327787	0.646136	0.606929
4	0.465958	0.880470	0.819825
5	0.563911	0.955723	0.871854

4.2 Part (b)

Table 5: Hold-to-maturity excess bond return regressions

	$H = 2$	$H = 3$	$H = 4$	$H = 5$
$\widehat{b}^{(H)}$ [HH t -stat]	0.6931 [3.4373]	0.5201 [2.9990]	0.3935 [2.2761]	0.3140 [2.0716]
R^2	8.0851%	5.4333%	3.7891%	2.8126%
N	811	811	811	811

4.3 Part (c)

I estimate Equation (4.2) separately for $H = 2, 3, 4, 5$ using monthly observations. I match the forward spread observed in month t with the excess annual bond return ending 12 months later. The regressions include an intercept, and the reported t -stats use Newey and West (1987, 1994) standard errors with an automatically selected lag length.

In Table 6, the estimated coefficients are positive and statistically significant at the 5% level for all four maturities. Thus, higher forward spreads are associated with higher excess bond returns over the following year. The predictive relationship is strongest for the four-year bond, which has the largest coefficient and the highest R^2 , although the coefficient falls somewhat at the five-year maturity.

Compared with the yield-spread regressions in Part (b), the forward-spread regressions have higher R^2 values for the three-, four-, and five-year bonds. This suggests that forward spreads contain more predictive information about one-year-ahead excess returns at these maturities. For $H = 2$, the coefficient and R^2 are the same as in Part (b), consistent with the relationship $xf_{b,t}^{(2)} = 2xy_{b,t}^{(2)}$.

Table 6: One-year excess bond return regressions on forward spreads

	2 years	3 years	4 years	5 years
<i>Forwards</i>				
$\hat{b}^{(H)}$ [NW t -stat]	0.6774 [3.2062]	0.8887 [3.3257]	1.1163 [3.5907]	0.9641 [2.9108]
R^2	7.6916%	8.6265%	10.7201%	6.6659%

4.4 Part (d)

I estimate the Cochrane–Piazzesi factor by regressing the average one-year-ahead excess return across the two- to five-year bonds on the current one- to five-year forward rates:

$$\frac{1}{4} \sum_{H=2}^5 x r_{b,t+1}^{(H)} = \theta_0 + \sum_{h=1}^5 \theta_h f_{b,t}^{(h)} + u_t.$$

Using the estimated slope coefficients, I construct

$$cp_t = \sum_{h=1}^5 \hat{\theta}_h f_{b,t}^{(h)}.$$

Figure 14 plots the estimated factor in percentage terms, with NBER recessions shown by the shaded areas. The CP factor varies considerably over time, suggesting substantial time variation in expected excess bond returns. Although the factor sometimes moves sharply around recessions, it does not follow the same pattern in every recession. The factor reaches its highest levels during the first half of the 1980s and remains relatively elevated in the early 1990s and early 2000s. It is comparatively low around 2020 and becomes negative during parts of 2022–2024.

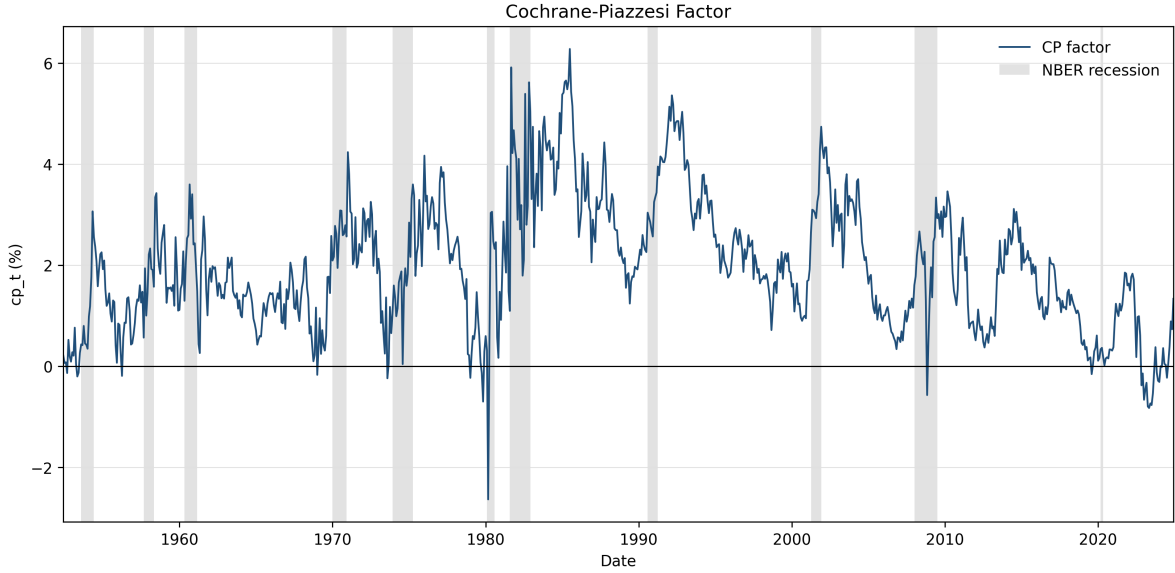


Figure 14: Cochrane–Piazzesi factor with NBER recession periods.

4.5 Part (e)

Table 7: One-year excess bond return regressions on the CP factor

	2 years	3 years	4 years	5 years
<i>CP factor</i>				
$\hat{b}^{(H)}$ [NW <i>t</i> -stat]	0.4418 [4.1257]	0.8274 [4.1243]	1.2517 [4.3896]	1.4791 [4.1986]
R^2	13.6703%	14.3774%	17.0468%	15.5228%